ASSETS 2026 Doctoral Consortium

Disability-Compensatory Generative AI Use:

From Restored Dignity to the Guardrail Paradox, Toward Disability-Inclusive AI Design

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What Motivated This Research

Sighted people access visual information instantly and independently; blind people depend on human intermediaries — a relational cost. GenAI promises to dissolve that asymmetry.

Yet for blind users, guardrails are not neutral — critical scholarship suggests this is no accident. They refuse legitimate disability-compensatory requests as privacy or sensitivity risks; once disability is disclosed or recognized, responses turn paternalistic, pitying, and distorted.

1. Both failures have been reported, but their structural mechanism has not been theorized — nor, to my knowledge, has their coexistence within one system been analyzed as a single phenomenon.
2. Rarely framed beyond convenience — as matters of independence, privacy autonomy, and human dignity.

One Cohort, One Program of Inquiry

Three interlocking studies, one cohort of seven blind university students in Korea, one IRB approval spanning the full program — drawn from a single, layered data corpus.

Three 90–120 minute focus group interviews on an audio-only platform, plus supplementary asynchronous data. Accessible oral consent: a screen-reader-accessible information sheet three days prior, oral consent audio-recorded at each session.

Analysis: reflexive thematic analysis — integrating the first author's own insider perspective as a blind screen reader user with the advisor's methodological review.

Three Interlocking Studies

Study 1 — Disability-Compensatory GenAI Use (Published, Korean Journal of Social Welfare)

Grounded in Morris's AI-augmented accessibility, the compensation principle in Baltes and Baltes' SOC model, and Zimmerman's self-regulated learning. Operationalized Disability-Compensatory GenAI Use as “the agentive strategies through which disabled users actively compensate for the sensory, physical, and structural constraints imposed by inaccessible environments, thereby recovering the learning agency that those environments had displaced.”

Study 2 — Everyday Independence and Dignity (Under review)

Grounded in Sen and Nussbaum's capability approach, Zimmerman's empowerment theory, and Jacobson's dignity framework. Beyond academic settings, GenAI negated the relational cost of help-seeking and let users cross previously impassable thresholds — experiences that converged into a restored sense of human dignity.

Study 3 — The Guardrail Paradox by Disability Decontextualization (Under review)

Disability Decontextualization

Grounded in critical algorithm studies and critical disability studies, and information ethics — Nissenbaum's contextual integrity and Mathiesen's informational justice.

Disability Decontextualization: the failure of AI systems to recognize the distinctive context in which blind users request visual information, so that their requests are evaluated as if made by a sighted requester and misclassified as risk.

The same design decision commits two parallel violations: of contextual integrity, built into safety architecture itself; and of informational justice, foreclosing access across all four dimensions at once.

Findings: Four Themes

Theme 1 · Disability context under-incorporation: Guardrail-Forced Disclosure by Disability Decontextualization

Requests such as checking clothing for menstrual staining were blocked. Participants had turned to a non-judgmental AI to avoid exposing private matters, but the refusal forced them back to family and personal assistants — a wider privacy exposure than the guardrail was meant to prevent.

Theme 2 · Disability context over-incorporation: Algorithmic Othering

Disability context over-incorporation took two forms: attitudinal repositioning and output distortion. Korean participants found the subtle, pervasive shift toward condescending sympathy more troubling than explicit pity, and problematized output distortion through unnecessary disability-linked content.

Theme 3 · Information reliability: Multi-Layered Verification Burden

General hallucination, disability-centric memory bias, and distorted interpretation together imposed a multi-layered verification burden. Cut off from visual cross-checking, blind users must verify outputs through other tools and sighted confirmation, spending markedly more energy than sighted users.

The economic dimension — The multimodal functions essential to Disability-Compensatory GenAI Use sit behind paid subscriptions: an optional convenience for sighted users, but a precondition of access for blind users.

Theme 4 · Critical agency: Disabled Users as Architects of Inclusive AI Design

Personal practice — Reconfigured their environments through system settings, prompting conventions that secured control over disability disclosure, and strategic cross-use of multiple AI tools.

Structural critique — Problematized current design assumptions and demanded disability-verified access pathways, disability context-aware filtering, and a shift from guardrail-based blocking toward privacy-secure AI design itself.

In the language of equal rights, not special accommodation — naming the differential blocking of visual information as structural inequality rather than technical error.

One Root Condition, Two Failures

Both failures share one root condition: the structural exclusion of disabled people from the processes that shape AI systems.

Disability context under-incorporation encodes the absence of disability context from design

Safety logic built without disability perspectives — disability-compensatory requests register only as surface-level risk patterns.

Disability context over-incorporation encodes the presence of ableism in data

Training data and human feedback saturated with the tragedy model — once disability becomes visible, the system reproduces the social script of pity and protection, through attitudinal repositioning and output distortion.

Not simply loosening guardrails — that trades one harm for another. Not simply adding disability context awareness — that would risk amplifying disability context over-incorporation.

> **A key principle: user-controlled disability context — which participants like P4 were already enacting from below through dual-control disclosure strategies, using an "!" at the start of a prompt to activate disability context on their own terms.**

Envisioned Contributions

Theoretical — Names and theorizes the Guardrail Paradox by Disability Decontextualization, extending contextual integrity theory into disability contexts.

Empirical — The first sustained empirical program on disabled people's GenAI use in Korea, documenting phenomena not previously reported: Guardrail-Forced Disclosure, attitudinal repositioning, the multi-layered verification burden and its economic dimension, and dual-control disclosure strategies.

Design & Policy — A social work argument that Disability-Compensatory GenAI Use be recognized as a matter of informational justice and reasonable accommodation rather than discretionary convenience.

Questions for the Consortium

1. How should an integrative framework be structured — and positioned — so that it speaks simultaneously to disability studies, social work, and the ASSETS community's design orientation, and its interdisciplinary character becomes a strength rather than a liability?
2. Is a design-oriented extension — moving from critique to architecture — methodologically advisable for a social work dissertation, and if so, what methods does the accessibility community consider rigorous for it?

Participant Voices

One participant quote for each theme, quoted verbatim.

> ### Theme 1 · Disability context under-incorporation: Guardrail-Forced Disclosure by Disability Decontextualization
>
> “At the end of the day, having to ask people around me for help actually exposes my private information even more. Many of us turn to AI precisely because we don't feel comfortable sharing those things with others.”
>
> — P6, Education major, totally blind

Theme 2 · Disability context over-incorporation: Algorithmic Othering

“For most everyday tasks, the kind of conversation where my visual impairment doesn't need to come into it at all. But when I do need it to take my visual impairment into account, I set it up so that if I start my prompt with an exclamation mark, it recognizes that as a signal that this is a request where my visual impairment is relevant.”

— P4, Special Education major, totally blind

Theme 3 · Information reliability: Multi-Layered Verification Burden

“Since there are always issues with information reliability, I end up having to do my own secondary research to double-check everything. To be honest, having to pay for a subscription while still carrying that extra burden of verification — it's a bit overwhelming and makes me question if it's really worth it.”

— P2, Korean Literature major, totally blind

Theme 4 · Critical agency: Disabled Users as Architects of Inclusive AI Design

“Actually, this is just a matter of basic rights. Sighted people can see these things instantly without any barriers or special devices… But for us, these restrictions exist simply because we have a visual impairment. That's exactly why I believe it's only right and fair for us to receive this kind of assistance through AI.”

— P7, Social Welfare major, totally blind

The Research Program

Study 1 (published)

Min, Sukhee, and Ji Sun Lee. 2026. “A Study on Generative AI Use Experiences of University Students with Visual Impairments: Focusing on GenAI-Augmented Accessibility in Academic Performance.” [in Korean] Korean Journal of Social Welfare 78 (1): 7–38.

Study 2 (under review)

Lee, Ji Sun, and Sukhee Min. 2026. “How Generative AI Restructures Everyday Independence and Dignity for People with Visual Impairments.” (Under review).

Study 2 is also the subject of my e-poster for the European Public Health Conference 2026. Accessible EPH 2026 page, with a 3-minute video and transcript.

Study 3 (under review)

The Guardrail Paradox by Disability Decontextualization — the focus of this poster. The full argument, methodology, and findings are set out in my ASSETS 2026 Doctoral Consortium submission (author's version, PDF). It is a doctoral consortium research statement, not a formal publication.

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